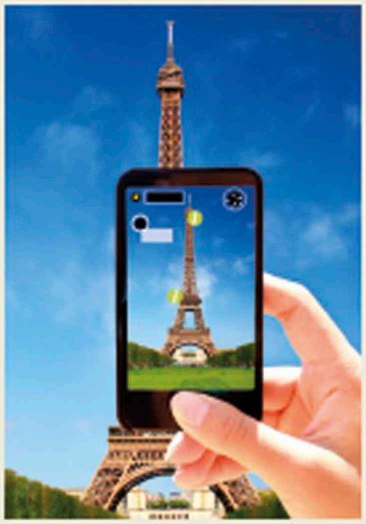
A connected world

Giant strides in computing and communications technology have enabled machines to share information, and as a result, billions of people all over the world are connected. Much of this now occurs via computer networks, which link computers and digital devices such as smartphones, allowing people to communicate with one another. Early networks were wired, needing physical connections, but wireless networks are now common.

The Internet
The largest network of linked computers on Earth, the Internet has grown in speed, availability, and number of useful applications, from email and web browsing to social media and live streaming of sound and video. Today, Internet signals are ferried by fiber optic data cables, by satellites, by telephone land lines, and wirelessly. This map shows Internet connections between cities in 2011, with the brightest areas having the most connections.



This smartphone is using an app to display facts about the landmarks being viewed.

Smartphones
These are powerful portable computing platforms that run sophisticated applications (apps). Smartphone apps enable many tasks to be performed on the move, from GPS mapping and video streaming to music editing and foreign language translation.

Going wireless

WiFi
WiFi enables billions of devices to connect to the Internet wirelessly, within a set range via a computer network, which usually requires a password to access.

RFID
Radio-frequency identification (RFID) uses radio waves to track objects, reading information stored on tiny RFID tags attached to an item.

Bluetooth
Bluetooth enables communication over short distances using ultra high frequency (UHF) radio waves. Devices may use bluetooth to connect to wireless printers or headphones.

NFC
Near field communications (NFC) allows contactless payments and other exchanges of data when digital devices are placed close to each other.

Key events

1969
The first messages were sent between two US computers over ARPANET, a precursor to the Internet. By 1972, 24 hosts (computer systems connected to the network) were on ARPANET, including NASA.

1974
American computer scientist Vinton Cerf and electrical engineer Bob Kahn developed Transmission Control Protocol (TCP), a set of rules that allow computers to send packets of data to each other over the Internet.

1990
The first Internet search engine, called Archie, was created by Canadian computer programmers. It searched through the indexes of FTP (File Transfer Protocol) sites looking for specific files.

1996
The Nokia 9000 Communicator was released. It was one of the first mobile phones with Internet access, enabling web browsing and email.

Nokia 9000 Communicator